MOVIE RECOMMENDATION SYSTEM USING MACHINE LEARNING

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1. Introduction

In the modern digital world, users are exposed to a vast number of movies across streaming platforms. Choosing relevant content from such large datasets is challenging. Recommendation systems play a crucial role in solving this problem by suggesting movies based on similarity and user preferences.

This project focuses on developing a **content-based movie recommendation system** using machine learning and natural language processing techniques.

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2. Problem Statement

Users often struggle to find movies that match their interests due to the overwhelming number of available options.

Goal:

To build a system that recommends movies similar to a selected movie based on content features such as genre, cast, keywords, and storyline.

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3. Objectives

- To build a content-based recommendation system
- To apply NLP techniques for feature extraction
- To improve recommendation accuracy using feature weighting
- To compute similarity between movies
- To develop an interactive web application

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4. Dataset Description

The dataset used is the **TMDB 5000 Movies Dataset**.

Features include:
- Movie title
- Overview (plot summary)
- Genres
- Keywords
- Cast
- Crew

Two datasets were merged:
- tmdb_5000_movies.csv

• tmdb_5000_credits.csv

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5. System Architecture

The system follows a machine learning pipeline:

1. Data Collection
2. Data Preprocessing
3. Feature Engineering
4. Text Processing
5. Vectorization
6. Similarity Computation
7. Recommendation
8. Deployment

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6. Data Preprocessing

• Merged datasets using movie title
• Removed missing values
• Removed duplicate records
• Converted JSON-like columns into lists using Python
• Extracted important features (genres, keywords, cast, crew)

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7. Feature Engineering

All features were combined into a single column called tags.

To improve performance, weights were assigned:
• Genres → High importance (×3)
• Overview → Moderate importance (×2)
• Keywords → Moderate importance (×2)
• Cast → Moderate importance (×2)
• Crew → Low importance (×1)

This ensures that more important features influence recommendations more strongly.

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8. Text Processing

• Converted text to lowercase
• Removed spaces between words
• Applied stemming using PorterStemmer
• Cleaned and normalized text

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9. Model Development

Vectorization

Used TF-IDF Vectorizer to convert text into numerical vectors.

Similarity Calculation

Used Cosine Similarity to measure similarity between movie vectors.

Recommendation Logic

• Find index of selected movie
• Retrieve similarity scores
• Sort movies based on similarity
• Return top 5 similar movies

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10. Deployment

The model was deployed using **Streamlit**, providing:
• Interactive UI
• Dropdown movie selection
• Real-time recommendations
• Movie posters using TMDB API

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12. Results and Insights

The system successfully recommends movies with similar themes and genres.

Example:
Input: Avatar

Output:
• Battle: Los Angeles
• Falcon Rising

• Aliens vs Predator: Requiem
• Aliens
• Lifeforce

The recommendations are relevant and consistent with the input movie.

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13. Challenges Faced

• Handling JSON-like structured data
• Feature extraction and cleaning
• Assigning appropriate weights
• Fixing indexing errors
• Integrating external API for posters

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14. Limitations

• Does not consider user preferences
• Limited to content-based filtering
• Dataset size constraints
• Cannot recommend trending/popular movies

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15. Future Scope

• Hybrid recommendation system
• Deep learning models (BERT embeddings)
• Genre and rating filters
• Cloud deployment
• User-based recommendation system

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16. Conclusion

This project demonstrates the practical application of machine learning and NLP in building a recommendation system. The use of **TF-IDF and weighted feature engineering** improves the accuracy and relevance of recommendations.
The system is efficient, scalable, and provides a user-friendly interface for real-time recommendations.